

## Arrays Concept

### Learning Objectives

- Explain what an array is and why it is useful for storing multiple values
- Access and modify elements in an array using index positions
- Describe zero-based indexing and how it applies to arrays

### Engage (5-7 minutes)

Imagine storing marks for 40 students in separate variables—mark1, mark2, all the way to mark40. This is impractical and makes it impossible to loop through them. An array solves this by storing all values in a single organized structure, like numbered lockers where each locker holds one item. Today we learn how arrays work and how to use them effectively.

### Explore (10-12 minutes)

Create an array of five favorite fruits. Access the first fruit, then the last fruit. What happens if you try to access an index that does not exist, like index 10? Now add a new fruit to the end and remove one from the beginning. Write down the index numbers and the values at each step to understand how indexing works.

### Explain (10-12 minutes)

An array (or list) is a collection of elements stored in sequential memory locations, each identified by an index. Most programming languages use zero-based indexing, meaning the first element is at index 0, the second at index 1, and so on. Arrays have a fixed or dynamic size, and you can access, modify, add, or remove elements using their index positions. Arrays are fundamental because they allow you to process collections of data using loops.

### Elaborate (8-10 minutes)

Create an array of 10 integers. Write code to find the sum, average, maximum, and minimum values using loops. Then reverse the array without using built-in reverse functions—swap elements from opposite ends moving inward. Discuss with a partner: what are the time complexity implications of accessing an element at the beginning versus the end of an array?

### Evaluate (5-8 minutes)

Given an array [23, 45, 12, 67, 89, 34], write the output of each operation: accessing index 2, updating index 4 to 99, inserting 56 at index 3, and removing index 0. Explain why zero-based indexing is used in most programming languages.